CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech.(EC) Examination

November 2013

EC 201 Digital Electronics & Logic Design

Date: 21.11.2013, Thursday | **Time: 01:30 p.m. To 04:30 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

Q - 1 (a) Convert the following numbers to decimal: (1) $(12121)_3$ (2) $(50)_7$ (3) $(8.3)_9$ (4) $(198)_{12}$ [04]

(b) Find the 10's Compliment of $(935)_{11}$. [01]

(c) Define following terms:(1) Power Dissipation (2) Propogation Delay [02]

Q - 2 (a) Convert the following functions in sum of minterms and product of maxterms. [07]

(1) $F(A,B,C) = (A'+B)(B'+C)$

(2) $F(A,B,C) = (AB+C)(B+AC)$

(3) $F(A,B,C) = 1$

(b) Simplify the Boolean function F using don't care conditions d, in (1) sum of products (2) product of sums. [07]

$F(A,B,C,D) = \sum (1,3,7,11,15)$ and $d(A,B,C,D)= \sum (0,2,5)$

OR

Q - 2 (a) Simplify the following Boolean function by means of the tabulation method [07]

$F(A,B,C,D) = \sum (0,1,2,8,10,11,14,15)$

(b) Design a combinational circuit that generates the 9's complement of a BCD digit. [07]

Q - 3 (a) Obtain NAND logic diagram of a following boolean function: [04]

$C= XY + YZ + XZ$

(b) Explain Magnitude Comparator in detail. [05]

(c) Implement the following function with a multiplexer: [05]

$F(A,B,C,D)= \sum(0,2,4,6,8,9,11,14,15)$

OR

Q - 3 (a) Draw the K-Map for a Four variable (1) Exclusive-OR function and (2) Equivalence Function. [04]

(b) Design a combinational circuit using a ROM. The circuit accepts a 3-bit number and generates an output binary number equal to the square of the input number. [05]

(c) A combinational Circuit defined by the functions. [05]

$F_1 = \sum(3,5,6,7)$ and $F_2 = \sum(0,2,4,7)$

Implement the circuit with Programmable Logic Array(PLA).

SECTION – II

Q - 4 (a) Given the Boolean function: [04]

$F = AB + A'B' + B'C$

(1) Implement it with only OR and NOT gates.

(2) Implement it with only AND and NOT gates.

(b) Show the Logic Diagram,Characteristic Table and Characteristic Equation of clocked RS flip-flop. [03]

Q - 5 (a) Explain Operation of the D-type edge-triggred flip-flop. [07]

(b) Design a counter that counts the decimal digits according to 2,4,2,1 code.Use T flip-flops. [07]

OR

Q - 5 (a) Design a sequencial circuit with four flip-flops A,B,C and D. The next states of B,C and D are equal to the present states of A,B and C respectively.The next state of A is equal to the exclusive-OR of the present states of C and D. [07]

(b) Design a counter with the following sequence:0, 1, 2, 3, 4 and repeat. Use JK flip-flops. [07]

Q - 6 (a) Define Register.Explain 4-bit register with parallel load. [07]

(b) Explain BCD Ripple Counter with necessary figure. [07]

OR

Q - 6 (a) Design 4 bit binary Up-Down Counter with necessary figure. [07]

(b) Explain Memory Unit with necessary Block Diagram.. [07]